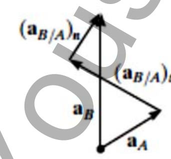
Relative Motion Analysis: Acceleration

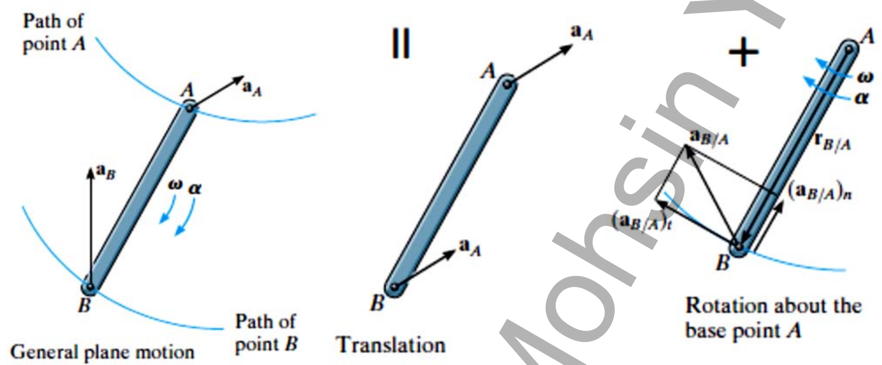
Position, Displacement, & Velocity

- Equation that relates acceleration of two points on a rigid bar,

$$\frac{dv_B}{dt} = \frac{dv_A}{dt} + \frac{dv_{B|A}}{dt} \text{ becomes } \boldsymbol{a}_B = \boldsymbol{a}_A + \boldsymbol{a}_{B|A}$$

$$\boxed{\boldsymbol{a}_B = \boldsymbol{a}_A + \left(\boldsymbol{a}_{B|A}\right)_t + \left(\boldsymbol{a}_{B|A}\right)_n}$$





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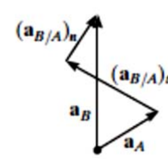


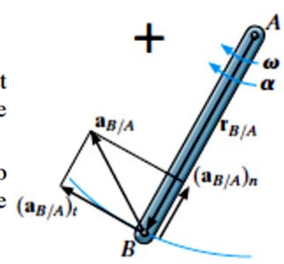
$\mathbf{a}_B$ = acceleration of point B

$\mathbf{a}_A$ = acceleration of point A

$(\mathbf{a}_{B/A})_t$ = tangential acceleration component of B with respect to A. The *magnitude* is $(a_{B/A})_t = \alpha r_{B/A}$, and the *direction* is perpendicular to $\mathbf{r}_{B/A}$.

$(\mathbf{a}_{B/A})_n$ = normal acceleration component of B with respect to A. The *magnitude* is $(a_{B/A})_n = \omega^2 r_{B/A}$, and the *direction* is always from B towards A.

$$\boxed{\boldsymbol{a}_B = \boldsymbol{a}_A + \boldsymbol{\alpha} \times \boldsymbol{r}_{B|A} - \omega^2 \boldsymbol{r}_{B|A}}$$



Rotation about the base point A

Procedure for Analysis: Relative Acceleration

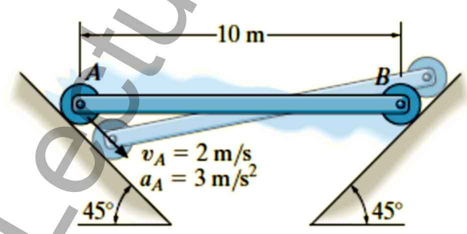
Velocity & Vector Analysis

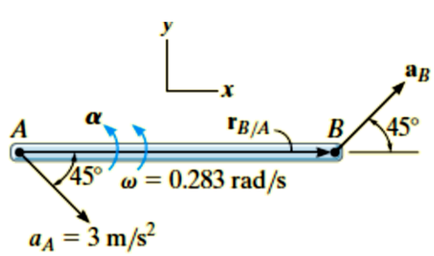
- **Velocity Analysis:**
- Determine the angular velocity $\boldsymbol{\omega}$ of the body using velocity analysis as already done in the previous section. Also determine the velocities $\boldsymbol{v}_A$ and $\boldsymbol{v}_B$ of points A and B if these points move along curved paths.
- **Kinematic Diagram:**
- Establish the directions of the fixed x, y coordinates and draw a kinematic diagram of the body. Indicate on it $\boldsymbol{a}_A$, $\boldsymbol{a}_B$, $\boldsymbol{\omega}$, $\boldsymbol{\alpha}$ and $\boldsymbol{r}_{B|A}$.
- If points A and B move along curved paths, then their accelerations should be indicated in terms of their tangential and normal components, i.e.,
- $\boldsymbol{a}_A = (\boldsymbol{a}_A)_t + (\boldsymbol{a}_A)_n$ and $\boldsymbol{a}_B = (\boldsymbol{a}_B)_t + (\boldsymbol{a}_B)_n$
- **Acceleration Equation:**
- Represent all the vectors in Cartesian vector notation,

$$\boldsymbol{a}_B = \boldsymbol{a}_A + \boldsymbol{\alpha} \times \boldsymbol{r}_{B|A} - \omega^2 \boldsymbol{r}_{B|A}$$

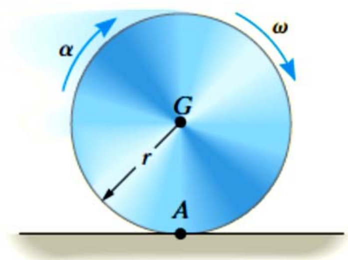
The rod AB shown in Fig. 16–27a is confined to move along the inclined planes at A and B. If point A has an acceleration of 3 m/s^2 and a velocity of 2 m/s, both directed down the plane at the instant the rod is horizontal, determine the angular acceleration of the rod at this instant.

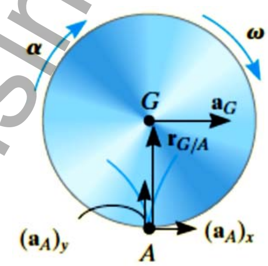
$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$



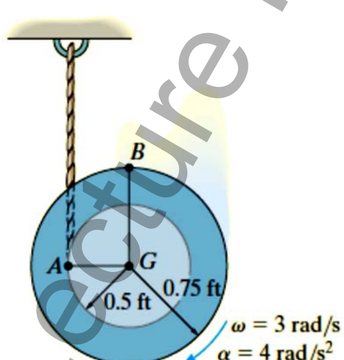


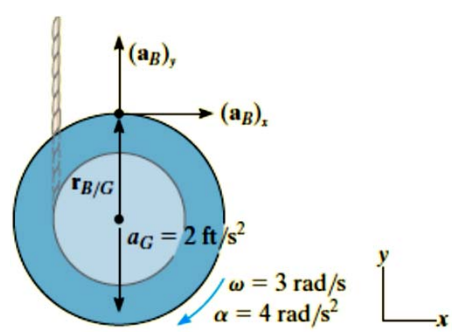
At a given instant, the cylinder of radius r, shown in Fig. 16–28a, has an angular velocity $\boldsymbol{\omega}$ and angular acceleration $\boldsymbol{\alpha}$. Determine the velocity and acceleration of its center G and the acceleration of the contact point at A if it rolls without slipping.





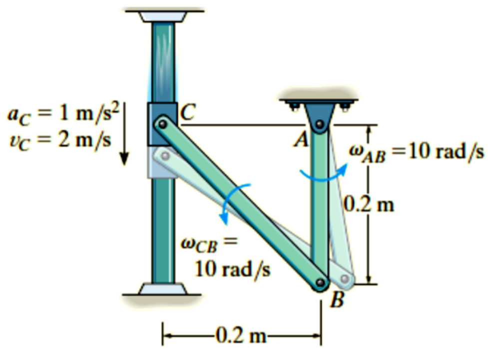
The spool shown in Fig. 16–29a unravels from the cord, such that at the instant shown it has an angular velocity of 3 rad/s and an angular acceleration of 4 rad/s^2. Determine the acceleration of point B.





The collar C in Fig. 16–30a moves downward with an acceleration of 1 m/s^2. At the instant shown, it has a speed of 2 m/s which gives links CB and AB an angular velocity $\omega_{AB} = \omega_{CB} = 10$ rad/s. (See Example 16.8.) Determine the angular accelerations of CB and AB at this instant.

$$\mathbf{v}_B = \mathbf{v}_C + \boldsymbol{\omega}_{CB} \times \mathbf{r}_{B/C}$$



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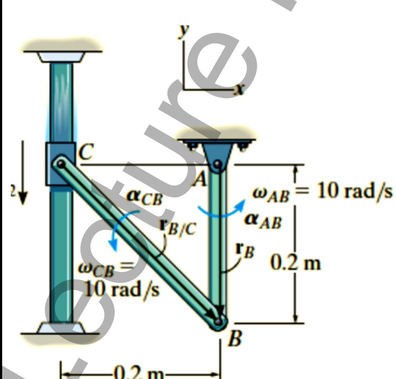
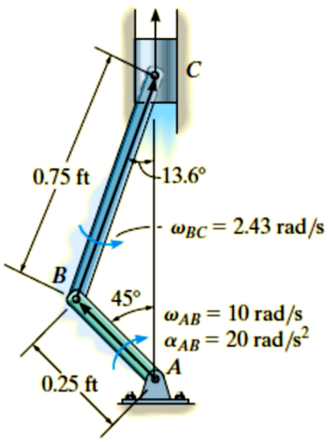


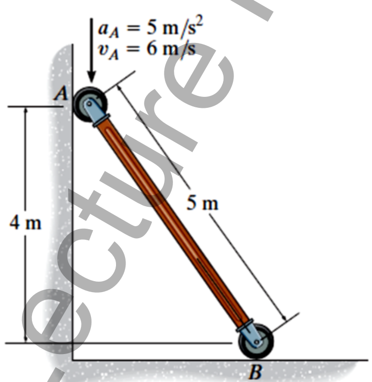
The crankshaft AB turns with a clockwise angular acceleration of 20 rad/s^2, Fig. 16–31*a*. Determine the acceleration of the piston at the instant AB is in the position shown. At this instant ω_{AB} = 10 rad/s and ω_{BC} = 2.43 rad/s (See Example 16.13.)



F16–19. At the instant shown, end A of the rod has the velocity and acceleration shown. Determine the angular acceleration of the rod and acceleration of end B of the rod.

$$\omega = \frac{v_A}{r_{A/IC}} = \frac{6}{3} = 2 \text{ rad/s}$$

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$



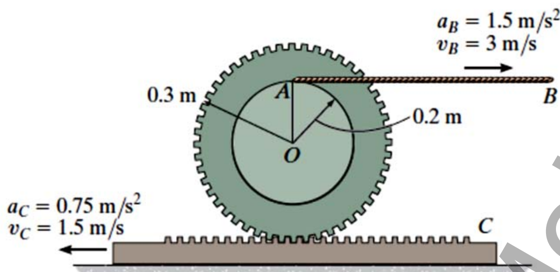
$$a_B = 4\alpha - 12$$
$$0 = 3\alpha + 11$$
$$\alpha = -3.67 \text{ rad/s}^2 \quad \textit{Ans.}$$
$$a_B = -26.7 \text{ m/s}^2 \quad \textit{Ans.}$$

F16–22. At the instant shown, cable AB has a velocity of 3 m/s and acceleration of 1.5 m/s^2, while the gear rack has a velocity of 1.5 m/s and acceleration of 0.75 m/s^2. Determine the angular acceleration of the gear at this instant.

$$\frac{r_{A/IC}}{3} = \frac{0.5 - r_{A/IC}}{1.5}; \qquad r_{A/IC} = 0.3333\text{ m}$$

$$\omega = \frac{v_A}{r_{A/IC}} = \frac{3}{0.3333} = 9\text{ rad/s}$$

$$\mathbf{a}_A = \mathbf{a}_C + \boldsymbol{\alpha} \times \mathbf{r}_{A/C} - \omega^2 \mathbf{r}_{A/C}$$



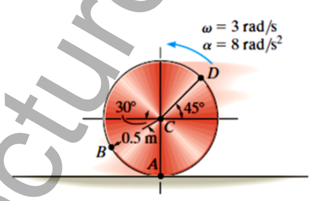
$$1.5 = 0.5\alpha - 0.75$$
$$\alpha = 4.5\text{ rad/s}^2$$

Relative Motion Analysis: Acceleration

Practice Problems

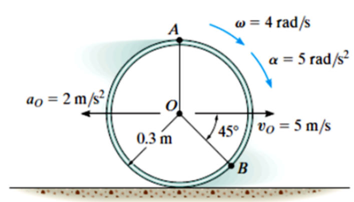
•16–109. The disk is moving to the left such that it has an angular acceleration $\alpha = 8$ rad/s^2 and angular velocity $\omega = 3$ rad/s at the instant shown. If it does not slip at A, determine the acceleration of point B.

16–110. The disk is moving to the left such that it has an angular acceleration $\alpha = 8$ rad/s^2 and angular velocity $\omega = 3$ rad/s at the instant shown. If it does not slip at A, determine the acceleration of point D.



16–111. The hoop is cast on the rough surface such that it has an angular velocity $\omega = 4$ rad/s and an angular acceleration $\alpha = 5$ rad/s^2. Also, its center has a velocity $v_O = 5$ m/s and a deceleration $a_O = 2$ m/s^2. Determine the acceleration of point A at this instant.

***16–112.** The hoop is cast on the rough surface such that it has an angular velocity $\omega = 4$ rad/s and an angular acceleration $\alpha = 5$ rad/s^2. Also, its center has a velocity of $v_O = 5$ m/s and a deceleration $a_O = 2$ m/s^2. Determine the acceleration of point B at this instant.

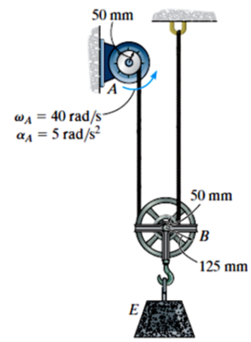


Relative Motion Analysis: Acceleration

PRACTICE PROBLEMS

16–123. Pulley A rotates with the angular velocity and angular acceleration shown. Determine the angular acceleration of pulley B at the instant shown.

***16–124.** Pulley A rotates with the angular velocity and angular acceleration shown. Determine the acceleration of block E at the instant shown.



16–130. Gear A is held fixed, and arm DE rotates clockwise with an angular velocity of $\omega_{DE} = 6$ rad/s and an angular acceleration of $\alpha_{DE} = 3$ rad/s^2. Determine the angular acceleration of gear B at the instant shown.

16–131. Gear A rotates counterclockwise with a constant angular velocity of $\omega_A = 10$ rad/s, while arm DE rotates clockwise with an angular velocity of $\omega_{DE} = 6$ rad/s and an angular acceleration of $\alpha_{DE} = 3$ rad/s^2. Determine the angular acceleration of gear B at the instant shown.